Bharat Kaurav

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Professional Summary

Software and AI Engineer with track record of architecting high-scale enterprise platforms and optimizing generative AI pipelines, including a 5x latency reduction for clinical diagnostics and $300k+ ARR growth for YC-backed startups

Work Experience

- **AI Engineer - Contractor** *Dec 2025 – Present*
  Qure.ai – Research & Optimization Team Bangalore, India (Remote)
  - **Redesigned** a sequential 5-stage AI pipeline into an **orchestrated parallel architecture**, implementing **model tiering** for generating Logic Trees to reduce recommendation generation latency from **26s to <5s** and remove LLM API costs
  - **Developed** a high-scale **evaluation framework** to validate clinical accuracy (Lung-RADS, Fleischner) across **10k+ test cases**, establishing a repeatable automated testing loop to measure and maintain **100%+ pass rates** during iterations
- **Software Engineer - Contractor** *Dec 2025 – Present*
  Foundation Industries (YC X25) – Full Stack & DevOps Team San Francisco, USA (Remote)
  - **Architected** an enterprise manufacturing platform with **$300k ARR**, high-performance **React Three Fiber** 3D CAD viewer, robust **multi-file upload pipeline** via **Vercel Blob** for 50MB+ datasets, integrating **Stripe, Supabase** for automated workflows
  - **Engineered** high-availability infrastructure, reducing bundle sizes by implementing **dynamic imports, lazy loading** for heavy libraries; streamlined operations by building automated notification layer using **React Email**, **Resend**, and **Slack Webhooks**
- **Software Engineer** *May 2025 – Oct 2025*
  Metzev Inc. Georgia, USA (Remote)
  - **Spearheaded the end-to-end development** of an EV battery platform for **15k+ users, multiple enterprise clients**, achieving **$50k ARR** by architecting a scalable **Stripe/Supabase** infrastructure
- **MLOps Engineer** *Nov 2024 – Apr 2025*
  unstudio.ai – Research & Backend Development Team Gurgaon, India (Remote)
  - Deployed a **FLUX personalization pipeline** using **LoRA, ControlNets** and **Redis-backed distributed system**, achieving **35% higher image quality**, **60% faster training/inference**, for European furniture clients
- **Machine Learning Consultant** *Jun 2024 – Oct 2024*
  Styldod Inc. – Research & Optimization Team Bangalore, India (Remote)
  - Spearheaded diffusion model acceleration via OneDiff & StableFast and developed an automated CV pipeline, achieving **50% faster inference**, a **30% GPU reduction**, and a **75% decrease** in manual effort across **300k+ images**
- **Projects & Collaborations**
  - **Virtual Try-On Optimization (Alle, Dec 2024)**: Directed 5-engineer team optimizing LoRA-based FLUX system, delivering **2.5× faster startup** and **98.7 PSNR** on A40/H100 GPUs
  - **LLM-SQL Generation (Attentions.ai, Apr 2024)**: Led 7-member team fine-tuning LLaMA, Gemma, Qwen & Phi models for enterprise SQL tasks, achieving **~90% accuracy** on complex benchmarks

Projects

- **Dynamic Agentic RAG system** *Dec 2024*
  13th Inter IIT-Tech Meet, IIT Bombay Github
  - Architected production-grade **multi-pipeline RAG system** with **3 specialized architectures** (HybridFL, FLARE, DRAGIN) supporting both legal & financial domains with **real-time knowledge graph updates** via Pathway
  - Achieved **15-20% performance gains** over GPT-4 and open-source LLMs on custom evaluation datasets; implemented **Docker-containerized microservices** with separate vector stores and **Streamlit production UI**
- **Behavior Simulation Challenge** *Nov 2023*
  12th Inter IIT-Tech Meet, IIT Madras Github
  - **Bronze Medal (Rank 3/22+ IITs)** - Engineered dual-task ML pipeline: tweet engagement prediction model achieving **96% accuracy** and content generation system using fine-tuned **LLaMA architecture**
  - Developed scalable **multimodal data processing pipeline** handling **1M+ heterogeneous records** (text, image, audio, video) with optimized feature extraction and model training workflows

Open Source Contributions

- **EchoMimicV2 & Float**: Developed notebooks to standardize initial model setup and workflow testing
- **PixArt Sigma**: Debugged and fixed errors in the LoRA training script, enabling fine-tuning of models

Publications

- **B. Kaurav**, S. S. Dar, A. Jain, C. S. Raghaw, M. Z. U. Rehman, N. Kumar. "An explainable deep neural network with frequency-aware channel and spatial refinement for flood prediction in sustainable cities." *Sustainable Cities and Society*, Vol. 130, 2025. https://doi.org/10.1016/j.scs.2025.106480
- **Under Review:** Forest-fire detection using a modified YOLO architecture; Landslide area segmentation with remote sensing imagery from Sentinel-2 satellite

Technical Skills

- **Languages & Tools:** Python, C/C++, SQL, Cursor, Claude Code, Shell Scripting
- **ML & Generative AI:** PyTorch, HuggingFace Transformers, Diffusers, LoRA, BitsAndBytes, LangChain/LlamaIndex
- **Web & Backend:** React, Next.js, FastAPI, Flask, Django, PostgreSQL, Stripe, Vercel
- **MLOps & Infrastructure:** Docker, AWS (EC2/S3), GitHub Actions, Redis, Supabase

Education & Achievements

- **B.Tech in Computer Science and Engineering**, Indian Institute of Technology Indore
  GPA: **8.66/10 (Current)** **Oct 2022 – May 2026**
- Inter IIT Tech Meet 12.0 Bronze Medal; JEE Advanced rank 1083 (out of 150k+ applicants); JEE Mains rank 5995 (out of 1M+ applicants); among top 300 candidates for IPhO India team